Experiment 1: Class Fundamentals

Program 1.1: Student Information System

```
class Student {
    int rollNo;
    String name;
    String branch;
    double cgpa;
    void display() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Branch: " + branch);
        System.out.println("CGPA: " + cgpa);
    }
}
public class StudentInfo {
    public static void main(String[] args) {
        Student s = new Student();
        s.rollNo = 101;
        s.name = "Raina";
        s.branch = "CSE-AIML";
        s.cgpa = 9.2;
        s.display();
    }
}
```

Output:

```
Roll Number: 101
Name: Raina
Branch: CSE-AIML
CGPA: 9.2
```

Program 1.2: Employee Information System

```
class Employee {
    int employeeId;
    String name;
    String department;
    double salary;
    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Department: " + department);
        System.out.println("Salary: " + salary);
    }
}
public class EmployeeInfo {
    public static void main(String[] args) {
        Employee e = new Employee();
        e.employeeId = 101;
        e.name = "Raina";
        e.department = "IT";
        e.salary = 50000;
        e.display();
    }
}
```

Output:

```
Employee ID: 101
Name: Raina
Department: IT
Salary: 50000.0
```

Experiment 2: Declaring Objects

Program 2.1: Book Details

```
class Book {
    String title;
    String author;
    double price;
    void display() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
        System.out.println();
    }
}
public class BookDetails {
    public static void main(String[] args) {
        Book b1 = new Book();
        Book b2 = new Book();
        Book b3 = new Book();
        b1.title = "Java Programming";
        b1.author = "James Gosling";
        b1.price = 500;
        b2.title = "Python Basics";
        b2.author = "John Smith";
        b2.price = 400;
        b3.title = "Data Structures";
        b3.author = "Mark Allen";
        b3.price = 600;
        b1.display();
        b2.display();
        b3.display();
```

```
    }
}
```

Output:

```
Title: Java Programming
Author: James Gosling
Price: 500.0
Title: Python Basics
Author: John Smith
Price: 400.0
Title: Data Structures
Author: Mark Allen
Price: 600.0
```

Program 2.2: Array of Student Objects

```
import java.util.Scanner;
class Student {
    int rollNo;
    String name;
    String branch;
    void display() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Branch: " + branch);
        System.out.println();
    }
}
public class StudentArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student[] students = new Student[5];
```

```
        for (int i = 0; i < 5; i++) {
            students[i] = new Student();
            System.out.println("Enter details of Student " + (i + 1));
            System.out.print("Roll Number: ");
            students[i].rollNo = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            students[i].name = sc.nextLine();
            System.out.print("Branch: ");
            students[i].branch = sc.nextLine();
        }
        System.out.println("\nStudent Details:");
        for (int i = 0; i < 5; i++) {
            students[i].display();
        }
        sc.close();
    }
}
```

Output:

```
Enter details of Student 1
Roll Number: 101
Name: Raina
Branch: CSE-AIML
Enter details of Student 2
Roll Number: 102
Name: Rinku
Branch: CSE
Enter details of Student 3
Roll Number: 103
Name: Ananya
```

Branch: CSE-AIML

Enter details of Student 4

Roll Number: 104

Name: Kiran

Branch: ECE

Enter details of Student 5

Roll Number: 105

Name: Priya

Branch: IT

Student Details:

Roll Number: 101

Name: Raina

Branch: CSE-AIML

Roll Number: 102

Name: Rinku

Branch: CSE

Roll Number: 103

Name: Ananya

Branch: CSE-AIML

Roll Number: 104

Name: Kiran

Branch: ECE

Roll Number: 105

Name: Priya

Branch: IT

Experiment 3: Assigning Object Reference Variables

Program 3.1: Reference Assignment

```
class Student {
    String name;
```

```
    void display() {
        System.out.println("Name: " + name);
    }
}
public class ReferenceAssignment {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.name = "Raina";
        Student s2 = s1;
        System.out.println("Before modification:");
        s2.display();
        s1.name = "Rinku";
        System.out.println("After modification:");
        s2.display();
    }
}
```

Output:

Before modification:

Name: Raina

After modification:

Name: Rinku

Program 3.2: Comparing Object References

```
class Employee {
    int id;
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
```

```
}
public class CompareReferences {
   public static void main(String[] args) {
      Employee e1 = new Employee(101, "Rahul");
      Employee e2 = e1;
      if (e1 == e2) {
         System.out.println("Both references point to the same object.");
      } else {
         System.out.println("References point to different objects.");
      }
   }
}
```

Output:

Both references point to the same object.

Experiment 4: Introducing Methods

Program 4.1: Calculator Using Methods

```
class Calculator {
   int add(int a, int b) {
      return a + b;
   }
   int subtract(int a, int b) {
      return a - b;
   }
   int multiply(int a, int b) {
      return a * b;
   }
   double divide(int a, int b) {
      return (double) a / b;
   }
```

```
}
public class CalculatorDemo {
    public static void main(String[] args) {
        Calculator c = new Calculator();
        int a = 20;
        int b = 10;
        System.out.println("Addition: " + c.add(a, b));
        System.out.println("Subtraction: " + c.subtract(a, b));
        System.out.println("Multiplication: " + c.multiply(a, b));
        System.out.println("Division: " + c.divide(a, b));
    }
}
```

Output:

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2.0

Program 4.2: Rectangle Operations

```
class Rectangle {
    double length;
    double width;
    double area() {
        return length * width;
    }
    double perimeter() {
        return 2 * (length + width);
    }
    void display() {
        System.out.println("Area: " + area());
        System.out.println("Perimeter: " + perimeter());
```